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https://github.com/TusharSamal3012/Housecorr3D-v2

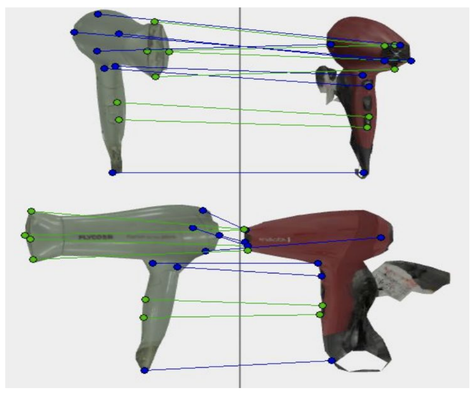
3D Correspondences via 2D Image Encoders

SMRITHI V SAMBREKAR, TUSHAR SAMAL, PRASHANTH PREMAKUMAR
University of Freiburg

LEONHARD SOMMER

Introduction

Problem: 3D semantic correspondence - automatically finding matching points between two 3D objects of the same category (e.g. two hair dryers), from shape and appearance alone, with no manual labeling.



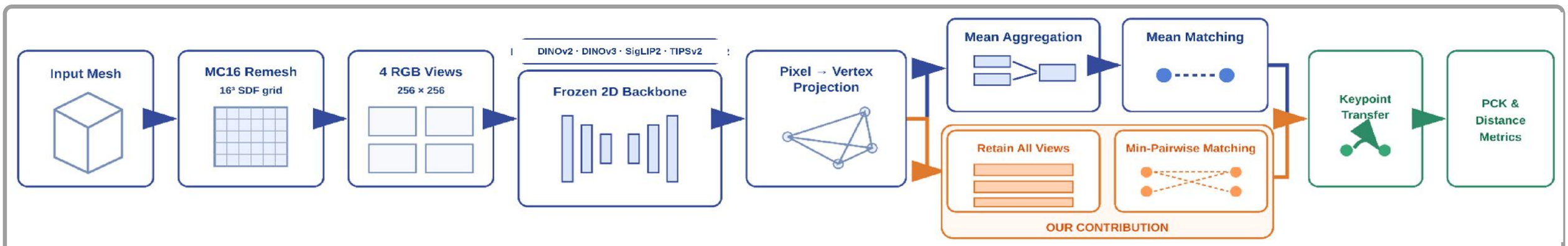
Open questions this project investigates:

1. Does averaging every viewpoint's feature into one mean descriptor throw away useful information?
2. Does the choice of 2D foundation-model backbone (self-supervised vs. image-text contrastive) affect correspondence quality?

Related work

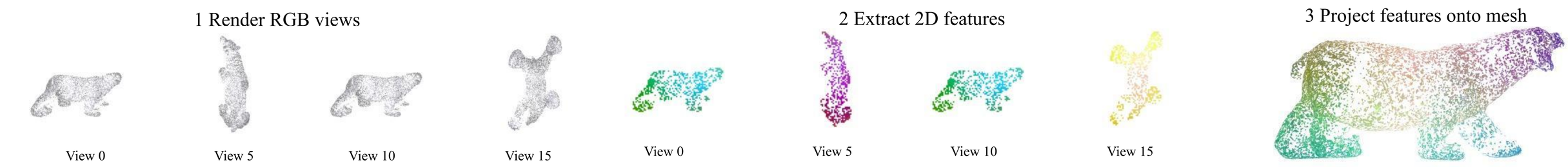
1. DINOv2 / DINOv3 - self-supervised, image-only (student-teacher self-distillation across augmented views, no text, no labels). Never compresses to a single global embedding, so patch-level features stay spatially precise historically the strongest backbone family for dense correspondence.
2. SigLIP2 - image-text contrastive (CLIP-style, but with a sigmoid rather than softmax loss). Optimizes global image-caption alignment; SigLIP2 adds a DINO-style self-distillation objective specifically to shore up its otherwise weaker dense features.
3. TIPSv2 - combines both: image-text contrastive learning and masked-image self-distillation in one model, aiming to get DINO-like dense features and SigLIP-like semantic alignment together rather than trading one off for the other.

Method



Pipeline Walkthrough

Finding: Multi-view 2D features remain spatially coherent after projection onto the mesh.



Quantitative Results

PCK@0.05 vs PCK@0.1 vs min-pairwise PCK@0.1

Encoder	PCK@0.05 ↑	PCK@0.1 ↑	Min-pairwise PCK@0.1
DINOv2-S	0.1436	0.2906	0.14962
DINOv2-B	0.1436	**0.3137**	0.14850
DINOv3-B	**0.1511**	0.2966	**0.27590**
SigLIP2	0.1282	0.2654	0.19392
TIPSv2	0.1425	0.2693	0.24910

Category wise results

Categories	Best Encoder	PCK@0.05	PCK@0.1	Euclidian distance
Dinosaur	DINOv2-S	0.1778	0.2961	0.4595
Hair Dryer	TIPSv2	0.2909	0.4816	0.4442
Teapot	SigLIP2	0.1269	0.3000	0.6454
Toy Animals	DINOv2-B	0.1319	0.2301	0.6100
Toy Plane	DINOv2-B	0.1473	0.2663	0.7046

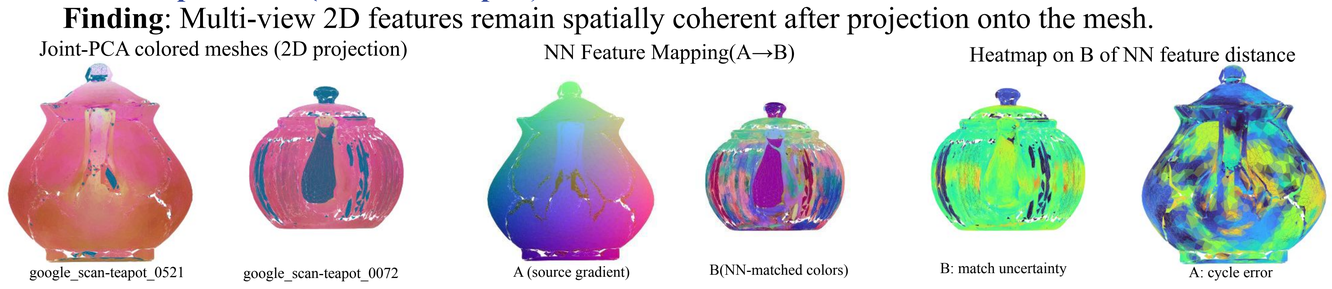
Key takeaways : DINOv3-B is the strongest overall encoder — it leads on PCK@0.05 (0.1511) and dominates min-pairwise PCK@0.1 (0.2759), well ahead of TIPSv2 (0.249) and SigLIP2 (0.194), though DINOv2-B edges it out on the looser PCK@0.1 metric (0.3137 vs 0.2966). However, no encoder wins universally at the category level: DINOv2-S is best for Dinosaur, TIPSv2 for Hair Dryer, SigLIP2 for Teapot, and DINOv2-B for Toy Animals and Toy Plane. Hair Dryer is the easiest category (highest PCK@0.1 at 0.4816) while Teapot is the hardest (largest Euclidean distance error at 0.6454) — suggesting encoder choice should ideally be category-aware rather than one-size-fits-all.

Benchmark: Built on HouseCorr3D, evaluated here as HouseCorr3Dv2. Dataset sourced from Omni6DPose: 153 categories / 4,302 instances in its metadata; 303 objects have the 3D keypoint annotations used for evaluation. This project's local setup covers 5 categories.

Qualitative Results

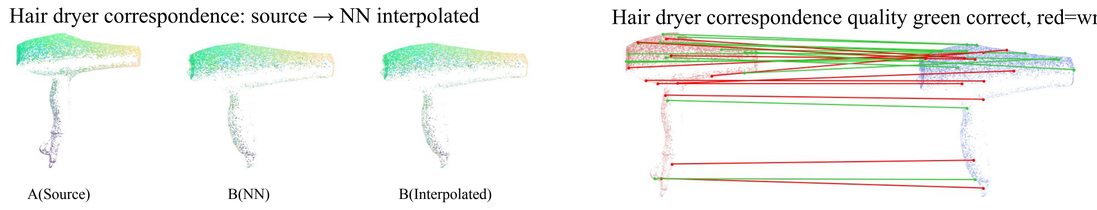
Dense Correspondence (Ceramic Teapot)

Finding: Multi-view 2D features remain spatially coherent after projection onto the mesh.



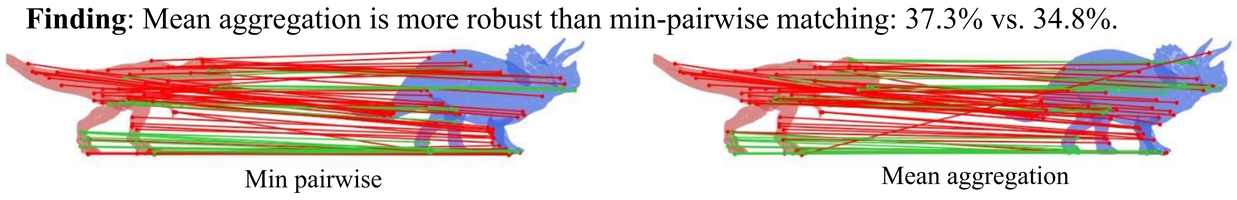
Category-Level Correspondence (Hair Dryer)

Finding: Feature matching transfers category-level structure, but only 56% of correspondences are correct.

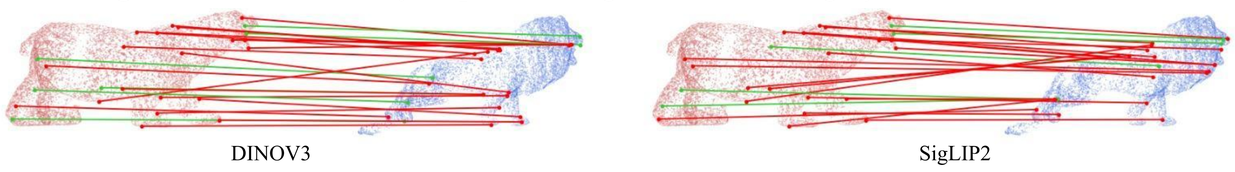


Dinov3 Results

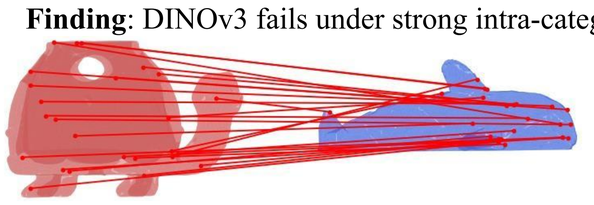
Finding: Mean aggregation is more robust than min-pairwise matching: 37.3% vs. 34.8%.



Finding: DINOv3 substantially outperforms SigLIP2 on this category: 28% vs. 9%.



Finding: DINOv3 fails under strong intra-category variation, reaching only 2.4%.



Acknowledgement: Images were generated by Codex

References : [1] Benchmarking Semantic Object Correspondence in Vision Foundation Models (arXiv:2605.31597)